

1.3 Matrices and Their Column Spaces

1 $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ is a **3 by 2 matrix**: $m = 3$ rows and $n = 2$ columns. Rank 2.

2 The 3 components of Ax are dot products of the 3 rows of A with the vector x :

$$\begin{array}{l} \text{Row at a time} \\ A \text{ times } x \end{array} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 2 \cdot 8 \\ 3 \cdot 7 + 4 \cdot 8 \\ 5 \cdot 7 + 6 \cdot 8 \end{bmatrix} = \begin{bmatrix} 23 \\ 53 \\ 83 \end{bmatrix}.$$

3 Ax is also a combination of the columns of A $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + 8 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$

4 The column space of A contains all combinations $Ax = x_1 a_1 + x_2 a_2$ of the columns.

5 **Rank one matrices**: All columns of A (and all combinations Ax) are on **one line**.

Sections 1.1 and 1.2 explained the mechanics of vectors—linear combinations, dot products, lengths, and angles. We have vectors in $\mathbf{R}^2$ and $\mathbf{R}^3$ and every $\mathbf{R}^n$.

Section 1.3 begins the algebra of m by n matrices: our true goal. A typical matrix A is a rectangle of m times n numbers— **m rows and n columns**. If m equals n then A is a “square matrix”. The examples below are 3 by 3 matrices.

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & -3 \\ 0 & 4 & 7 \\ 0 & 0 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 4 & 7 \\ -3 & 7 & 5 \end{bmatrix}$
Identity matrix	Diagonal matrix	Triangular matrix	Symmetric matrix

We often think of the columns of A as vectors $a_1, a_2, \dots, a_n$. Each of those n vectors is in m -dimensional space. In this example the a 's have $m = 3$ components each:

$$\begin{array}{l} m = 3 \text{ rows} \\ n = 4 \text{ columns} \\ \text{3 by 4 matrix } A \end{array} \quad A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

This is a “difference matrix” because A times x produces a vector Ax of differences like $x_2 - x_1$. *How does an m by n matrix A multiply an n by 1 vector x ?* There are two ways to the same answer—we work with the rows of A or we work with the columns.

The row picture of Ax will come from dot products of x with the rows of A .

The column picture will come from linear combinations of the columns of A .

Row picture of $A\mathbf{x}$ Each row of A multiplies the column vector $\mathbf{x}$. Those multiplications *row times column* are dot products! The first dot product comes from row 1 of A :

$$(\text{row } 1) \cdot \mathbf{x} = (-1, 1, 0, 0) \cdot (x_1, x_2, x_3, x_4) = x_2 - x_1.$$

It takes m times n small multiplications to find the $m = 3$ dot products that go into $A\mathbf{x}$.

Three rows
Three dot products

$$A\mathbf{x} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \text{row } 1 \cdot \mathbf{x} \\ \text{row } 2 \cdot \mathbf{x} \\ \text{row } 3 \cdot \mathbf{x} \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \end{bmatrix} \quad (1)$$

Notice well that each row of A has the same number of components as the vector $\mathbf{x}$. Four columns of A multiply x_1 to x_4 . Otherwise multiplying $A\mathbf{x}$ would be impossible.

Column picture of $A\mathbf{x}$ The matrix A times the vector $\mathbf{x}$ is a combination of the columns of A . The n columns are multiplied by the n numbers in $\mathbf{x}$. Then add those column vectors $x_1\mathbf{a}_1, \dots, x_n\mathbf{a}_n$ to find the same vector $A\mathbf{x}$ as in equation (1):

$$A\mathbf{x} = x_1(\text{column } \mathbf{a}_1) + x_2(\text{column } \mathbf{a}_2) + x_3(\text{column } \mathbf{a}_3) + x_4(\text{column } \mathbf{a}_4) \quad (2)$$

This combination of n columns involves exactly the same multiplications as dot products of $\mathbf{x}$ with the m rows. But it is higher level! We have a vector equation instead of three dot products. You see the same result $A\mathbf{x}$ in equation (1) above and equation (3) below.

Combination of columns

$$A\mathbf{x} = x_1 \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \end{bmatrix} \quad (3)$$

Let me admit something right away. If I have numbers in A and $\mathbf{x}$, and I want to compute $A\mathbf{x}$, then I tend to *use dot products*: the row picture. But if I want to *understand* $A\mathbf{x}$, the column picture is better. “The column vector $A\mathbf{x}$ is a combination of the columns of A .”

We are aiming for a picture of not just one combination $A\mathbf{x}$ of the columns (from a particular $\mathbf{x}$). What we really want is a picture of **all combinations of the columns** (from multiplying A by all vectors $\mathbf{x}$). This figure shows one combination $2\mathbf{a}_1 + \mathbf{a}_2$ and then it tries to show the plane of all combinations $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$ (for every x_1 and x_2).

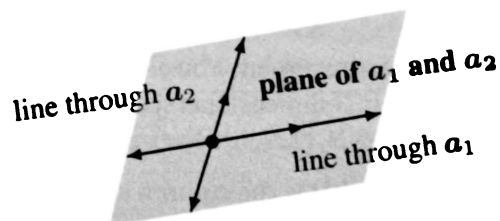
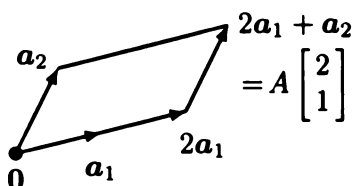


Figure 1.7: A linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$. All linear combinations fill a plane.

The next important words are **independence**, **dependence**, and **column space**.

Here is a key point! Columns of A might or might not contribute anything new. They might be combinations of earlier columns (which we already included). Example 1 shows 3 columns that give 3 new directions. Those columns are “independent”:

Example 1
Independent columns $A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}$ Each column gives a new direction.
 Their combinations fill 3D space $\mathbf{R}^3$.

If we look at all combinations of the columns, we see all vectors (b_1, b_2, b_3) in 3D space. The first column $x_1(1, 2, 3)$ allows us to match any number b_1 . Then $x_2(0, 4, 5)$ leaves b_1 alone and we can match any number b_2 . Finally $x_3(0, 0, 6)$ doesn't touch b_1 and b_2 and allows us to match any b_3 . We have found x_1, x_2, x_3 so that $A_1\mathbf{x} = \mathbf{b}$.

Independence means: The only combination of columns that produces $A\mathbf{x} = (0, 0, 0)$ is $\mathbf{x} = (0, 0, 0)$. The columns are **independent** when each new column is a vector that we don't already have as a combination of previous columns. Independence will be important.

Example 2
Dependent columns $A_2 = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ 6 & 0 & 6 \end{bmatrix}$ Column 1 + column 2 = column 3 $1 + 2 = 3$
 Their combinations don't fill 3D space $1 + 4 = 5$
 $6 + 0 = 6$

The opposite of independent is “*dependent*”. These three columns of A_2 are **dependent**. Column 3 is in the plane of columns 1 and 2. Nothing new from column 3.

I usually test independence going from left to right. The column $(1, 1, 6)$ is no problem. Column 2 is *not* a multiple of column 1 and $(2, 4, 0)$ gives a new direction. But column 3 is the sum of columns 1 and 2. The third column vector $(3, 5, 6)$ is not independent of $(1, 1, 6)$ and $(2, 4, 0)$. That matrix A_2 only has two independent columns.

If I went from right to left, I would start with independent columns 3 and 2. Then column 1 is a combination (column 3 minus column 2). Either way we find that the three columns are in the **same plane**: two independent columns produce a plane in 3D.

That plane is the column space of this matrix: Plane = all combinations of the columns.

Dependent columns in Example 2

column 1 + column 2 – column 3 is $(0, 0, 0)$.

Example 3 $A_3 = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 5 & 15 & 20 \end{bmatrix}$ Now $\mathbf{a}_2$ is 3 times $\mathbf{a}_1$. And $\mathbf{a}_3$ is 4 times $\mathbf{a}_1$.
Every pair of columns is dependent.

This example is important. You could call it an extreme case. All three columns of A_3 lie on the **same line** in 3-dimensional space. That line consists of all column vectors $(c, 2c, 5c)$ —all the multiples of $(1, 2, 5)$. Notice that $c = 0$ gives the center point $(0, 0, 0)$.

That line in 3D is the column space for this matrix A_3 . The line contains all vectors $A_3\mathbf{x}$. By allowing every vector $\mathbf{x}$, we fill in the column space of A_3 —and here we only filled one line. That is almost the smallest possible column space.

The column space $C(A)$ contains all vectors $A\mathbf{x}$: All combinations of the columns.

Thinking About the Column Space of A

“Vector spaces” are a central topic. Examples are coming unusually early. They give you a chance to see what linear algebra is about. The combinations of all columns produce the column space, but you only need independent columns. So we start with column 1, and go from left to right in identifying independent columns. Here are two examples A_4 and A_5 .

$$A_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_5 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

A_4 has four independent columns. For example, column 4 is not a combination of columns 1, 2, 3. There are no dependent columns in A_4 . Triangular matrices like A_4 are easy provided the main diagonal has no zeros. Here the diagonal is 1, 1, 1, 1.

A_5 is not so easy. Columns 1 and 2 and 3 are independent. The big question is whether column 4 is a combination of columns 1, 2, 3. To match the final 1 in column 4, that combination will have to start with column 1.

To cancel the 1 in the top left corner of A_5 , we need *minus* the second column. Then we need *plus* column 3 so that -1 and $+1$ in row 2 will also cancel. Now we see what is true about this matrix A_5 : Only 3 independent columns because

$$\text{Column 4 of } A_5 = \text{Column 1} - \text{Column 2} + \text{Column 3}. \quad (4)$$

The next step is to “visualize” the column space—all combinations of the four columns. That word is in quotes because the task may be impossible. I don’t think that drawing a 4-dimensional figure would help (possibly this is wrong). A_4 is a good place to start.

Do you see why $C(A_4)$ is the whole space $\mathbf{R}^4$? If we look to algebra, we see that every vector v in $\mathbf{R}^4$ is a combination of the columns. Here is the combination:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = (v_1 - v_2) \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + (v_2 - v_3) \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + (v_3 - v_4) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + (v_4) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad (5)$$

This says: Every v is in the column space. We solved the four equations $A_4 x = v$!

The short useful word “span” describes all the linear combinations of a set of vectors. So the span of the columns of A (independent or not) is the column space.

Geometrically, here is one way to look at A_4 . The first column $(1, 0, 0, 0)$ is responsible for a line in 4-dimensional space. That line contains every vector $(c_1, 0, 0, 0)$. The second column is responsible for another line, containing every vector $(c_2, c_2, 0, 0)$. If you add every vector $(c_1, 0, 0, 0)$ to every vector $(c_2, c_2, 0, 0)$, you get a 2-dimensional plane inside 4-dimensional space. That plane is the *span* of columns 1 and 2.

That was the first two columns. The main rule of linear algebra is *keep going*. The last two columns give two more directions in $\mathbf{R}^4$. The four columns are independent. At the end, equation (5) shows how every 4-dimensional vector is a combination of the four columns of A_4 . The column space of A_4 is all of $\mathbf{R}^4$.

If we attempt the same plan for the matrix A_5 , the first 3 columns cooperate. But column 4 of A_5 is a combination of columns 1, 2, 3. Those three columns combine to give a *three-dimensional subspace* inside $\mathbf{R}^4$. Column 4 happens to be *in that subspace*.

That three-dimensional subspace is the whole column space $C(A_5)$. We can only solve $A_5\mathbf{x} = \mathbf{v}$ when $\mathbf{v}$ is in $C(A_5)$. The matrix A_5 only has three independent columns.

I always write $C(A)$ for the column space of A . When A has m rows, the columns are vectors in m -dimensional space $\mathbf{R}^m$. The column space might fill all of $\mathbf{R}^m$ or it might not. For $m = 3$, here are all four possibilities for column spaces in 3-dimensional space :

1. The whole space $\mathbf{R}^3$	A has 3 independent columns
2. A plane in $\mathbf{R}^3$ going through $(0, 0, 0)$	2 independent columns
3. A line in $\mathbf{R}^3$ going through $(0, 0, 0)$	1 independent column
4. The single point $(0, 0, 0)$ in $\mathbf{R}^3$	A is a matrix of zeros !

Here are simple matrices to show those four possibilities for the column space $C(A)$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$C(A) = \mathbf{R}^3 = xyz \text{ space}$
 $C(A) = xy \text{ plane}$
 $C(A) = x \text{ axis}$
 $C(A) = \text{one point } (0, 0, 0)$

Author's note The words “column space” did not appear in Chapter 1 of the 5th edition. I thought the idea of a *space* was too important to come so soon. Now I think that the best way to understand such an important idea is to see it early and often. It is examples more than definitions that make ideas clear—in mathematics as in life.

Here is a succession of questions. With practice in the next section 1.4, you will find the keys to the answers. They give a real understanding of any matrix A .

1. How many columns of A are independent? That number r is the “**rank**” of A .
2. Which are the first r independent columns? They are a “**basis**” for the column space.
3. What combinations of those r basic columns produce the remaining $n - r$ columns?
4. Write any A as an m by r column matrix C times an r by n matrix R : $A = CR$.
5. (Amazing) The r rows of R are a basis for the **row space** of A : combinations of rows.

Section 1.4 will explain how to multiply those matrices C and R . The result is $A = CR$. C contains columns from A . Please notice that the rows of R *do not* come directly from A .

Matrices of Rank One

Now we come to the building blocks for all matrices. *Every matrix of rank r is the sum of r matrices of rank one.* For a rank one matrix, all column vectors lie along the same line. That one line through $(0, 0, 0)$ is the whole column space of the rank one matrix.

Example $A_6 = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 12 & -8 \\ 2 & 6 & -4 \end{bmatrix}$ has rank $r = 1$. All columns: same direction!

Columns 2 and 3 are multiples $3a_1$ and $-2a_1$ of the first column $a_1 = (1, 4, 2)$. The column space $C(A_6)$ is only a line containing all vectors $ca_1 = (c, 4c, 2c)$.

Here is a wonderful fact about any rank one matrix. You may have noticed the rows of A_6 . **All the rows are multiples of one row.** When the column space is a single line in m -dimensional space, the row space is a single line in n -dimensional space. All rows of this matrix A_6 are multiples of ____.

An example like A_6 raises a basic question. **If all columns are in the same direction, why does it happen that all rows are in the same direction?** To find an answer, look first at this 2 by 2 matrix. Column 2 is m times column 1:

$$A = \begin{bmatrix} a & ma \\ b & mb \end{bmatrix} \quad \text{Is row 2 a multiple of row 1?}$$

Yes! The second row (b, mb) is $\frac{b}{a}$ times the first row (a, ma) . If the column rank is 1, then the row rank is 1. To cover every possibility we have to check the case when $a = 0$. Then row 1 = $\begin{bmatrix} 0 & 0 \end{bmatrix} = 0$ times row 2.

Our 2 by 2 proof is complete. Let me look at this 3 by 3 matrix. Rank = 1 if $a \neq 0$.

$$A = \begin{bmatrix} a & ma & pa \\ b & mb & pb \\ c & mc & pc \end{bmatrix} \quad \begin{array}{l} \text{Column 2 is } m \text{ times column 1} \\ \text{Column 3 is } p \text{ times column 1} \\ \text{Rows 2 and 3 are } b/a \text{ and } c/a \text{ times row 1} \end{array}$$

This matrix does not have two independent columns. Is it the same for the rows of A ? Is row 2 in the same direction as row 1? *Yes.* Is row 3 in the same direction as row 1? *Yes.* **The rule still holds.** The row rank of this A is also 1 (equal to the column rank).

Now jump from rank one matrices to all matrices. At this point we could make a guess: It looks possible that **row rank equals column rank for every matrix.** If A has r independent columns, then A has r independent rows. A wonderful fact!

I believe that this is the first great theorem in linear algebra. So far we have only seen the case of rank one matrices. The next section 1.4 will explain matrix multiplication AB and lead us toward an understanding of “row rank = column rank” for all matrices.